



PERTANDINGAN WORLDSKILLS MALAYSIA KATEGORI BELIA (WSMB) TAHUN 2026

(IT NETWORK SYSTEM ADMINISTRATION)

MODUL C

NETWORKING ENVIRONMENT

PERINGKAT AKHIR

MASA : 6 JAM

NAMA: _____

No. KP: _____

**JABATAN PEMBANGUNAN KEMAHIRAN
KEMENTERIAN SUMBER MANUSIA**

Contents

Contents	2
Introduction to Test Project	3
Introduction	3
Description of project and tasks.....	3
Basic Configuration.....	4
Switching.....	4
Routing.....	5
Services	7
Security	8
WAN and VPN	9
Network Address Table	10
IPv4 Routing Diagrams.....	13
Network Topology.....	14

Introduction to Test Project

The following is a list of sections or information that must be included in all Test Project proposals that are submitted to WorldSkills Malaysia Belia (WSMB).

- Contents including list of all documents, drawings and photographs that make up the Test Project
- Introduction/overview
- Short description of project and tasks
- Instructions to the Competitor
- Marking scheme (incl. assessment criteria)
- Other

Introduction

Network technology knowledge is becoming essential nowadays for people who want to build a successful career in any IT engineering field. This test project contains a lot of challenges from real life experience, primarily IT integration and IT outsourcing. If you are able to complete this project with the high score, you are definitely ready to service the network infrastructure for any multi-branch enterprise.

The competition has a fixed start and finish time. You must decide how to best divide your time. Please **carefully read** the following instructions!

When the competition time ends, please leave your station in a running state. The assessment will be done in the state as it is. ***No reboot will be initiated as well as powered off machines will not be powered on!***

Set all user password and credential with **Skills39** unless being specifically stated with password. Please use the information below for all the servers and clients.

Description of project and tasks

This test project is designed using a variety of network technologies that should be familiar from the Cisco certification tracks (CCNA). Tasks are broken down into following configuration sections:

- Basic configuration
- Switching
- Routing
- Services
- Security
- WAN and VPN

All sections are independent but all together they build very complex network infrastructure. Some tasks are pretty simple and straightforward; others may be tricky. You may see that some technologies are expected to work on top of other technologies. For example, IPv6 routing is expected to run on top of configured VPNs, which are, in turn, expected to run on top of IPv4 routing, which is, in turn, expected to run on top of PPPoE, and so on. It is important to understand that if you are unable to come up with a solution in the middle of such technology stack it doesn't mean that the rest of your work will not be graded at all. For example, you may not configure IPv4 routing that is required for VPN because of IP reachability but you can use static routes and then continue to work with VPN configuration and everything that runs on top. You won't receive points for IPv4 routing in this case but you will receive points for everything that you made operational on top as long as functional testing is successful.

Basic Configuration

1. Configure hostnames for all network devices as you see on the topology.
2. Configure domain name **wsmb2026.net** for all network devices on the topology.
3. Configure **Skills39** as a privileged mode password for all devices.
 - a. Only **PBKDF2** hash of the password should be stored in configuration for switch and router.
4. Configure IPv4/IPv6 address for all network devices as you see on the topology.
5. IPv6 for end devices should be obtained automatically through SLAAC. Manual setting is allowed.
6. Configure **MYT +8** as time zone for all network devices.
7. For your convenience, there are two important basic configurations needed to be done on the console
 - a. no ip domain lookup
 - b. exec-timeout 0 0 (on line console 0)

Switching

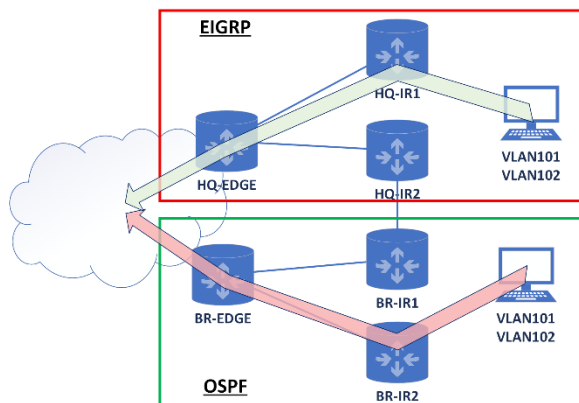
1. Configure VTP v3 on HQ switches to synchronize VLANs. It should be possible to modify VLAN database only from HQ-DSW1, and VLAN databases of other switches in HQ should be synchronized from HQ-DSW1. Use the domain name above with password **Skills39**. Assign the interface access port as required.
 - a. VLAN 101 with name HQSERVER
 - b. VLAN 102 with name HQCLIENT
 - c. VLAN 999 with name HQMGMT
2. Configure VTP v3 on BR switches to synchronize VLANs. It should be possible to modify VLAN database only from BR-DSW2, and VLAN databases of other switches in BR should be synchronized from BR-DSW2. Use the domain name above with password **Skills39**. Assign the interface access port as required.
 - a. VLAN 101 with name BRVL101
 - b. VLAN 102 with name BRVL102
 - c. VLAN 999 with name BRMGMT
3. Configure all links between switches as trunk port.
 - a. Do not use dynamic negotiation protocol. Explicitly **disable DTP**
 - b. Configure manual pruning so that only created VLANs are allowed forwarding.
 - c. Configure so that MGMT traffic should be forwarded **without IEEE 802.1q tag**.
4. Configure EtherChannel
 - a. All link between HQ-DSW1 and HQ-DSW2 with interface number 12
 - i. Use open IEEE standard as negotiation protocol between HQ-DSW1 and HQ-DSW2.
 - ii. HQ-DSW1 should initiate negotiation and HQ-DSW2 should respond but don't initiate.
 - iii. Configure the load balancing and forwarding method with source and destination MAC address.
 - b. All link between BR-DSW1 and BR-DSW2 with interface number 34
 - i. Use Cisco proprietary protocol as negotiation protocol between BR-DSW1 and BR-DSW2.
 - ii. BR-DSW2 should initiate negotiation and BR-DSW1 should respond but don't initiate.
 - iii. Configure the load balancing and forwarding method with source and destination IP address.
5. Configure STP on HQ network
 - a. The STP convergence is required to happen as quickly as possible for all network.
 - b. HQ-DSW1 should be STP root bridge for HQ VLAN. If HQ-DSW1 goes down, HQ-DSW2 should take over as the STP root bridge.
 - c. BR-DSW2 should be STP root bridge for BR VLAN. If BR-DSW2 goes down, BR-DSW1 should take over as the STP root bridge.
 - d. Configure port which is connected to end device so that it immediately begins forwarding when connected.

Routing

1. Configure **EIGRP** protocol in HQ network (HQ-EDGE, HQ-IR1, HQ-IR2, HQ-DSW1, HQ-DSW2). EIGRP is also used between devices involved in DMVPN (REM1-RTR, REM2-RTR).
 - a. Configure EIGRP process where IPv4 and IPv6 is enabled under single instance with name of "WSMB2026"
 - b. By default, all EIGRP Hello messages should be suppressed. Only enable it on interfaces where EIGRP adjacencies are required.
 - c. Configure EIGRP on IPv4
 - i. Use the autonomous system of 2026 for IPv4.
 - ii. Configure so that the Loopback1 address is the Router-ID.
 - iii. Include all of the internal subnets only available including Loopback1 in the BRANCH network.
 - iv. BR-EDGE should advertise default route to EIGRP only if it receives default route from ISP.
 - v. Make sure that routes from external routes are preferred through HQ-IR1. Routing table should only have one next-hop towards HQ-IR1 for HQ-DSW1 and HQ-DSW2. DO NOT change any properties on the GigabitEthernet interfaces (speed, delay etc)
 - vi. EIGRP should also advertise external routes that are learned from other protocols (BGP, OSPF).
 - vii. Make sure that devices in DMVPN only receive summarized route of 172.20.0.0/16, 172.21.0.0/16 and 172.30.0.0/16.
 - d. Configure EIGRP on IPv6.
 - i. Use the same instance WSMB2026 configured above with same AS number of 2026 for IPv6.
2. Configure **OSPF** protocol in BR network (BR-EDGE, BR-IR1, BR-IR2, BR-DSW1, BR-DSW2) and relevant devices (HQ-IR2)
 - a. Configure OSPF process 104 in each of the devices.
 - b. Use Loopback1 address as the Router-ID in each of the devices.
 - c. By default, all OSPF Hello messages should be suppressed. Only enable it on interfaces where OSPF adjacencies are required. Switches should obtain adjacencies through MGMT VLAN only.
 - d. BR-EDGE should advertise default route to OSPF only if it received default route from ISP.
 - e. Make sure that routes from external routes are preferred through HQ-IR1. Routing table should only have one next-hop towards HQ-IR1 for HQ-DSW1 and HQ-DSW2. DO NOT change any properties on the GigabitEthernet interfaces (speed, delay etc)
 - f. Summarize the network of backbone area so that HQ-IR2 only receive summarized network of 172.21.0.0/24 from area 0. Set the administrative distance as 200 for OSPF.
 - g. Summarize the network of external route from HQ so that BR devices only receive summarized network of 172.20.0.0/24 within that subnet from external route through ASBR.
 - h. OSPF should also advertise external routes that are learned from other protocols (BGP, EIGRP)
 - i. Make sure that BR-DSW2 is the DR router and BR-DSW1 is the BDR router on area 10. Other router should not participate in the DR/BDR election.
 - j. Configure mutual route redistribution between EIGRP and OSPF on HQ-IR2.
 - i. When HQ route (EIGRP) goes into BR (OSPF), it should come with tag value of 34
 - ii. When BR route (OSPF) goes into HQ (EIGRP), it should come with tag value of 12
 - iii. Prevent any routing loops while performing route redistribution.
3. Configure **OSPFv3** protocol in BR network (BR-IR1, BR-DSW1, BR-DSW2) and relevant devices (HQ-IR2)
 - a. Configure OSPFv3 process 106 in each of the devices with all within single backbone area.
 - b. OSPFv3 should also advertise external routes that are learned from other protocols.
 - c. Configure mutual route redistribution between EIGRPv6 and OSPFv3 on HQ-IR2 and also BR-IR1.
 - d. Prevent any routing loops while performing route redistribution.

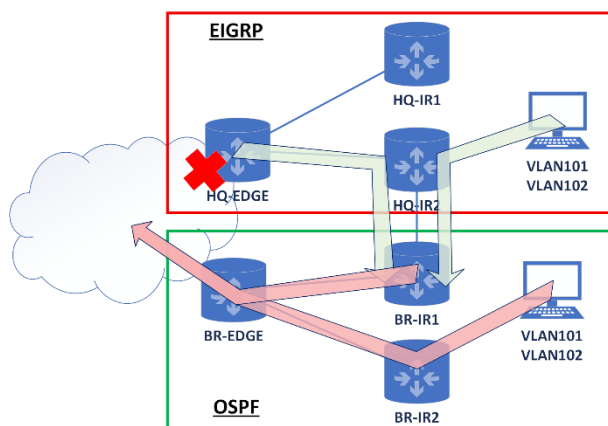
4. Configure BGP.
 - a. Configure eBGP session between edge devices with ISP router.
 - i. Make sure ISP router propagates default route to all other devices.
 - b. Advertise all networks of internet public network only into BGP. This includes the Loopback0 address on each router.
5. Mutual Redistribution Default Route
 - a. The default route should be obtained from the BGP as shown below

Normal Default Route



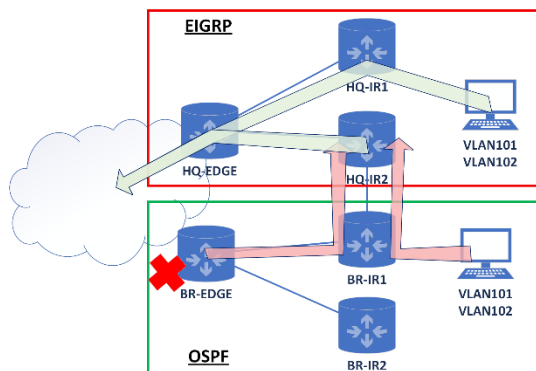
- b. If HQ-EDGE does not get default route from BGP, the default route should be redirected to BR-EDGE. Ensure it the default route goes back to BGP after the uplink is restored.

EIGRP failover to OSPF



- c. If BR-EDGE does not get default route from BGP, the default route should be redirected to HQ-EDGE. Ensure it the default route goes back to BGP after the uplink is restored.

OSPF Failover to EIGRP



Services

1. Configure NAT.
 - a. When devices in HQ network communicate with internet, these IP address should be translated to 98.76.1.50 – 98.76.1.60
 - b. When devices in BR network communicate with internet, these IP address should be translated to 98.76.1.100 – 98.76.1.110
 - c. When REM1 and REM2 communicate with internet, these IP address should be translated to IPv4 address of the interface which is connected to ISP on each router.
2. Configure DHCP for IPv4 network.
 - a. REM1-PC and REM2-PC can obtain IPv4 address automatically from their respective network.
 - a. REM1 and REM2 clients should use **WIN-SRV** as DNS server.
 - b. HQ network should be able to get DHCP from HQ-SRV. The DHCP is preconfigured
 - c. BR network should be able to get DHCP from BR-EDGE.
 - a. BR clients should use **HQ-SRV** as DNS server.
3. Configure FHRP on DSW1 and DSW2 on both BR and HQ switches for IPv4.
 - a. Make sure the higher priority device is always the take over as the preferred one.
 - b. Use Hot Standby Router Protocol v2 for VLAN 101.
 - i. DSW1 should be used as default gateway.
 - ii. Use 104 as group number of IPv4.
 - iii. The IP address should be the last IP address from the subnet.
 - c. Use a Hot Standby Router Protocol v2 for VLAN 102.
 - i. DSW2 should be used as default gateway.
 - ii. Use 204 as group number of IPv4.
 - iii. The IP address should be the last IP address from the subnet
4. Configure NTP to synchronize the clock
 - a. Set ISP as the main NTP server.
 - b. Edge devices should associate with ISP using their Loopback address.
 - c. Other internal L3 devices should set their respective edge device as the primary NTP server. Use Loopback address for NTP communication.
5. Configure network monitoring using SNMP on HQ-IR2 and BR-IR1.
 - a. Configure the device location as Putrajaya, Malaysia
 - b. Configure the system contact as admin@wsmb2026.net
 - c. Cacti monitoring server is already pre-configured on HQ-SRV. It should be accessible from <http://172.20.101.10/cacti> (username: admin, password: Skills39)
 - d. Use SNMPv3 authPriv security level with following paramters
 - i. SHA and AES128 using password: Skill39@MY
 - ii. Username of snmpuser with group of snmpgroup

Security

1. Configure console authentication on all HQ-EDGE and BR-EDGE.
 - a. Use local account. Create a maximum privilege user **admin** with password **Skills39**.
 - b. Login credentials should be obtained from RADIUS server from HQ-SRV with the pre-shared key as Skills39. Use Loopback address to connect to RADIUS server.
 - c. There are two users configured, radiusr1 (priv-lvl=15) and radiusr2 (priv-lvl=3) with password Skills39.
 - d. If RADIUS server does not respond, the local account should be used as fallback.
 - e. After successful authentication, users should automatically land in privileged mode based on privilege level pre-configured and included in the RADIUS server.
2. Configure remote access on ISP.
 - a. Make sure that only the most secure protocol is enabled for VTY. Disable plaintext communication.
 - b. Create a max privilege user named **sshuser**, password **Skills39**. Make sure this user able to login.
3. Configure port-security on the port which is connected to HQ-PC1 using following parameters:
 - a. Maximum MAC address – 2
 - b. In case of policy violation, security message should be displayed on the console, port should be disabled.
 - c. When device encounters any ErrDisable, reset the status after 3 minutes.

WAN and VPN

1. Certificate Authority is already pre-configured on WIN-SRV with Simple Certificate Enrolment Protocol.
 - a. Register and obtain RSA keys from WIN-SRV using FQDN with domain name of wsmb2026.net.
 - b. You can use manual entry of certificate or auto-enrolment SCEP, functional testing will be applied.
2. Configure tunnels between HQ-EDGE, REM1-RTR and REM2-RTR (DMVPN)
 - a. Remote sites should be able to communicate with HQ and also BRANCH sites through this tunnel.
 - b. Use **Tunnel 100** as VTI for all routers, with the source address using external Loopback address.
 - c. Use **Skills39** as authentication key
 - d. Make sure the tunnel is dual stack. Use network-id of 104 on IPv4 and 106 on IPv6.
 - e. Spoke-to-spoke traffic transmission should be possible
3. Configure IKEv2 VPN for VTI and Site-to-Site
 - a. Use the following parameters for tunnel VPN.
 - b. Phase 1 parameters:
 - a. Encryption: aes-cbc-256
 - b. Integrity: sha-256
 - c. DH Group: 14
 - d. PRF: sha-256
 - e. Authentication using RSA certificate from WIN-SRV
 - c. Phase 2 parameters:
 - a. Protocol: ESP
 - b. Encryption: aes-256
 - c. Hash: sha-256

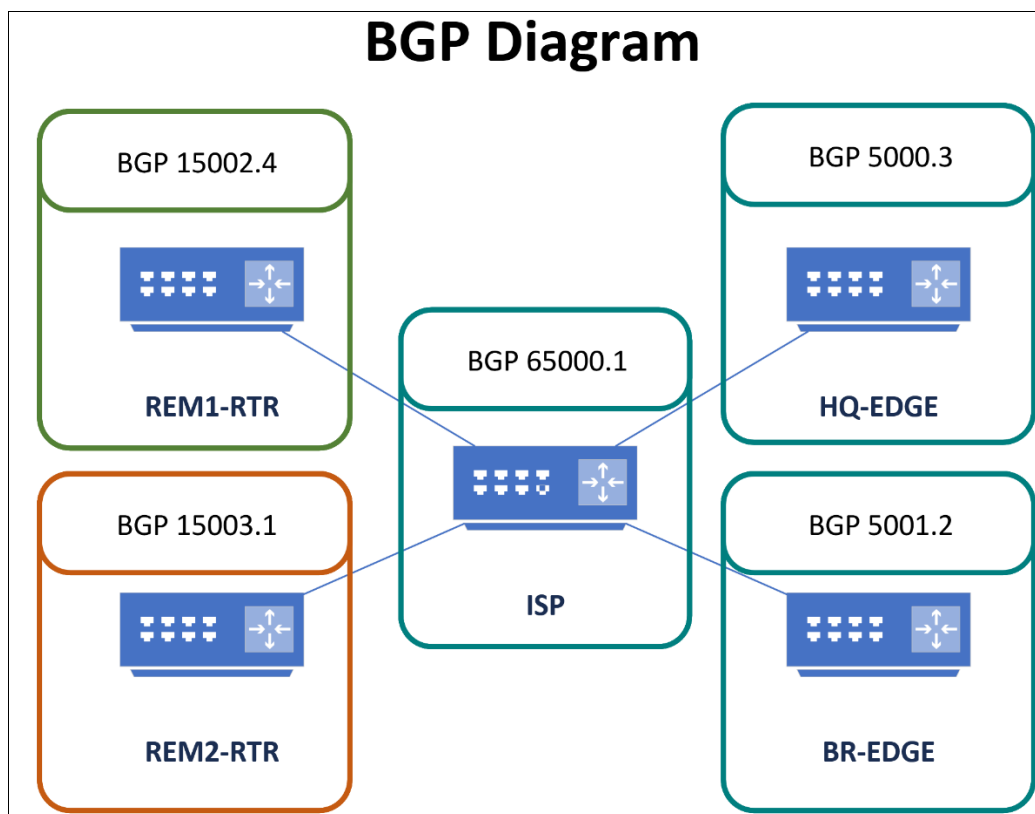
Network Address Table

Device	Interface	Address
ISP	Loopback0	8.8.8.8/32
	GigabitEthernet0/0	98.76.54.254/24
	GigabitEthernet0/1	98.76.1.62/26
	GigabitEthernet0/2	98.76.1.126/26
	GigabitEthernet0/3	98.76.2.254/24
	GigabitEthernet0/4	98.76.3.254/24
HQ-EDGE	Loopback0	1.1.1.1/32
	Loopback1	10.20.0.1/32
	GigabitEthernet0/0	98.76.1.1/26
	GigabitEthernet0/1	172.20.0.1/30
	GigabitEthernet0/2	172.20.0.5/30
	Tunnel100	172.31.1.1/24 dead:beef:cafe:ffff::1/64
HQ-IR1	Loopback0	10.20.0.2/32
	GigabitEthernet0/0	172.20.0.2/30
	GigabitEthernet0/1	172.20.0.9/30
	GigabitEthernet0/2	172.20.0.13/30
HQ-IR2	Loopback0	10.20.0.3/32
	GigabitEthernet0/0	172.20.0.6/30
	GigabitEthernet0/1	172.20.0.17/30
	GigabitEthernet0/2	172.20.0.21/30
	GigabitEthernet0/3	172.31.0.1/30
HQ-DSW1	Loopback0	10.20.0.4/32
	GigabitEthernet0/1	172.20.0.10/30
	GigabitEthernet0/2	172.20.0.22/30
	Vlan 101	172.20.101.251/24 dead:beef:cafe:1101::fff1/64
	Vlan 102	172.20.102.251/24 dead:beef:cafe:1102::fff1/64
	Vlan 999	172.20.99.251/24 dead:beef:cafe:1999::fff1/64
HQ-DSW2	Loopback0	10.20.0.5/32
	GigabitEthernet0/1	172.20.0.18/30
	GigabitEthernet0/2	172.20.0.14/30
	Vlan 101	172.20.101.252/24 dead:beef:cafe:1101::fff2/64
	Vlan 102	172.20.102.252/24 dead:beef:cafe:1102::fff2/64

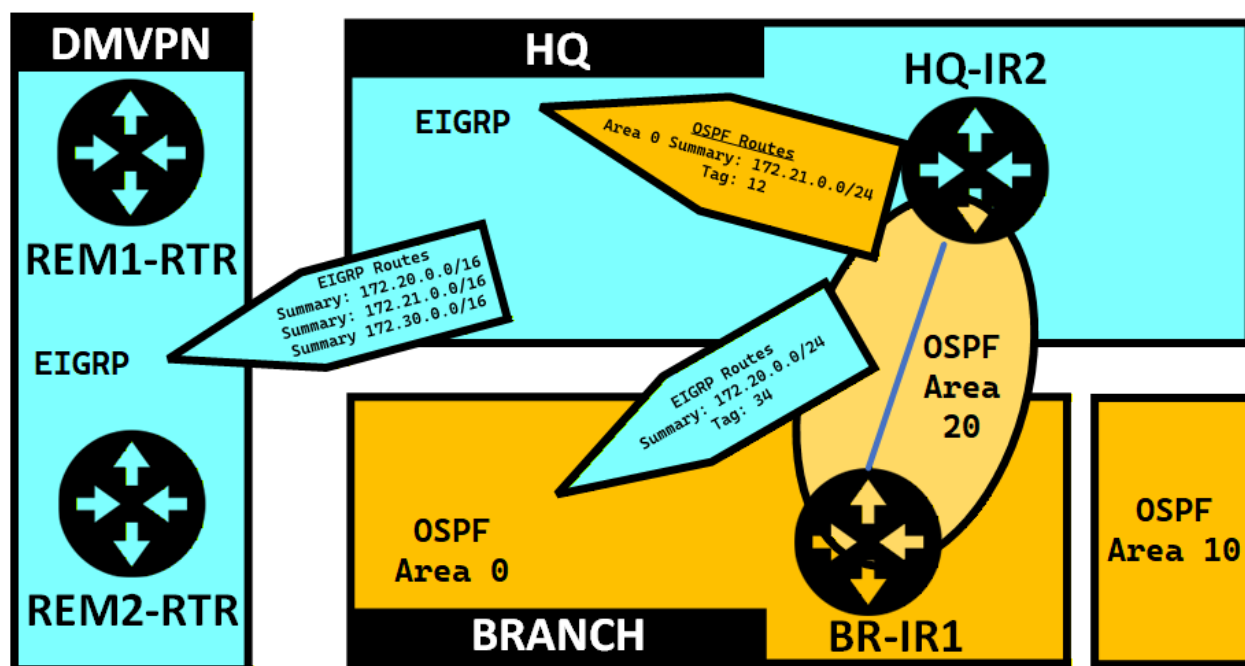
	Vlan 999	172.20.99.252/24 dead:beef:cafe:1999::fff2/64
HQ-ASW	Vlan 999	172.20.99.253/24 dead:beef:cafe:1999::fff3/64
BR-EDGE	Loopback0	2.2.2.2/32
	Loopback1	10.21.0.1/32
	GigabitEthernet0/0	98.76.1.65/26
	GigabitEthernet0/1	172.21.0.1/30
	GigabitEthernet0/2	172.21.0.5/30
BR-IR1	Loopback0	10.21.0.2/32
	GigabitEthernet0/0	172.21.0.2/30
	GigabitEthernet0/1	172.21.0.9/30
	GigabitEthernet0/2	172.21.0.13/30
	GigabitEthernet0/3	172.31.0.2/30
BR-IR2	Loopback0	10.21.0.3/32
	GigabitEthernet0/0	172.21.0.6/30
	GigabitEthernet0/1	172.21.0.17/30
	GigabitEthernet0/2	172.21.0.21/30
BR-DSW1	Loopback0	10.21.0.4/32
	GigabitEthernet0/1	172.21.0.10/30
	GigabitEthernet0/2	172.21.0.22/30
	Vlan 101	172.21.101.251/24 dead:beef:cafe:2101::fff1/64
	Vlan 102	172.21.102.251/24 dead:beef:cafe:2102::fff1/64
	Vlan 999	172.21.99.251/24 dead:beef:cafe:2999::fff1/64
BR-DSW2	Loopback0	10.21.0.5/32
	GigabitEthernet0/1	172.21.0.18/30
	GigabitEthernet0/2	172.21.0.14/30
	Vlan 101	172.21.101.252/24 dead:beef:cafe:2101::fff2/64
	Vlan 102	172.21.102.252/24 dead:beef:cafe:2102::fff2/64
	Vlan 999	172.21.99.252/24 dead:beef:cafe:2999::fff2/64
BR-ASW	Vlan 999	172.21.99.253/24 dead:beef:cafe:2999::fff3/64
REM1-RTR	Loopback0	3.3.3.3/32
	Loopback1	10.30.1.1/32 dead:beef:cafe::3333/128

	GigabitEthernet0/0	98.76.2.1/24
	GigabitEthernet0/1	172.30.10.254/24 dead:beef:cafe:3001::ffff/64
	Tunnel100	172.31.1.2/24 dead:beef:cafe:ffff::2/64
REM2-RTR	Loopback0	4.4.4.4/32
	Loopback1	10.30.1.2/32 dead:beef:cafe::4444/128
	GigabitEthernet0/0	98.76.3.1/24
	GigabitEthernet0/1	172.30.20.254/24 dead:beef:cafe:3002::ffff/64
	Tunnel100	172.31.1.3/24 dead:beef:cafe:ffff::3/64
WIN-SRV	Ethernet 0	98.76.54.10/24
HQ-SRV	Ethernet0	172.20.101.10/24 dead:beef:cafe:1101::10/64
HQ-PC1	Ethernet0	172.20.102.XXX/24 (DHCP) dead:beef:cafe:1102::X /64
HQ-PC2	Ethernet0	172.20.102.XXX/24 (DHCP) dead:beef:cafe:1102::X /64
BR-PC1	Ethernet0	172.21.101.XXX/24 (DHCP) dead:beef:cafe:2101::X /64
BR-PC2	Ethernet0	172.21.102.XXX/24 (DHCP) dead:beef:cafe:2102::X /64
REM1-PC	Ethernet0	172.30.10.XXX/24 (DHCP) dead:beef:cafe:3001::X /64
REM2-PC	Ethernet0	172.30.20.XXX/24 (DHCP) dead:beef:cafe:3002::X /64

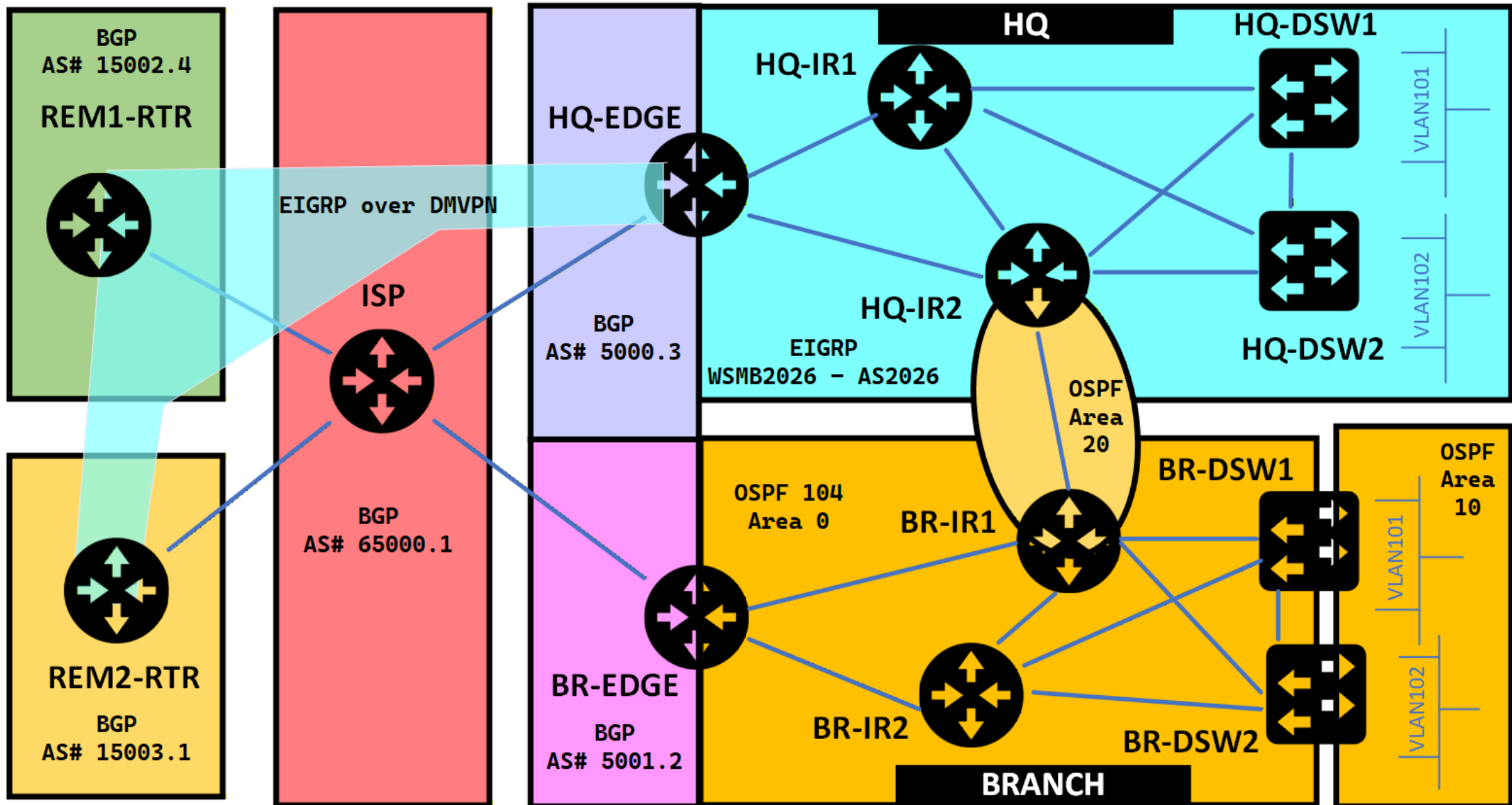
IPv4 Routing Diagrams



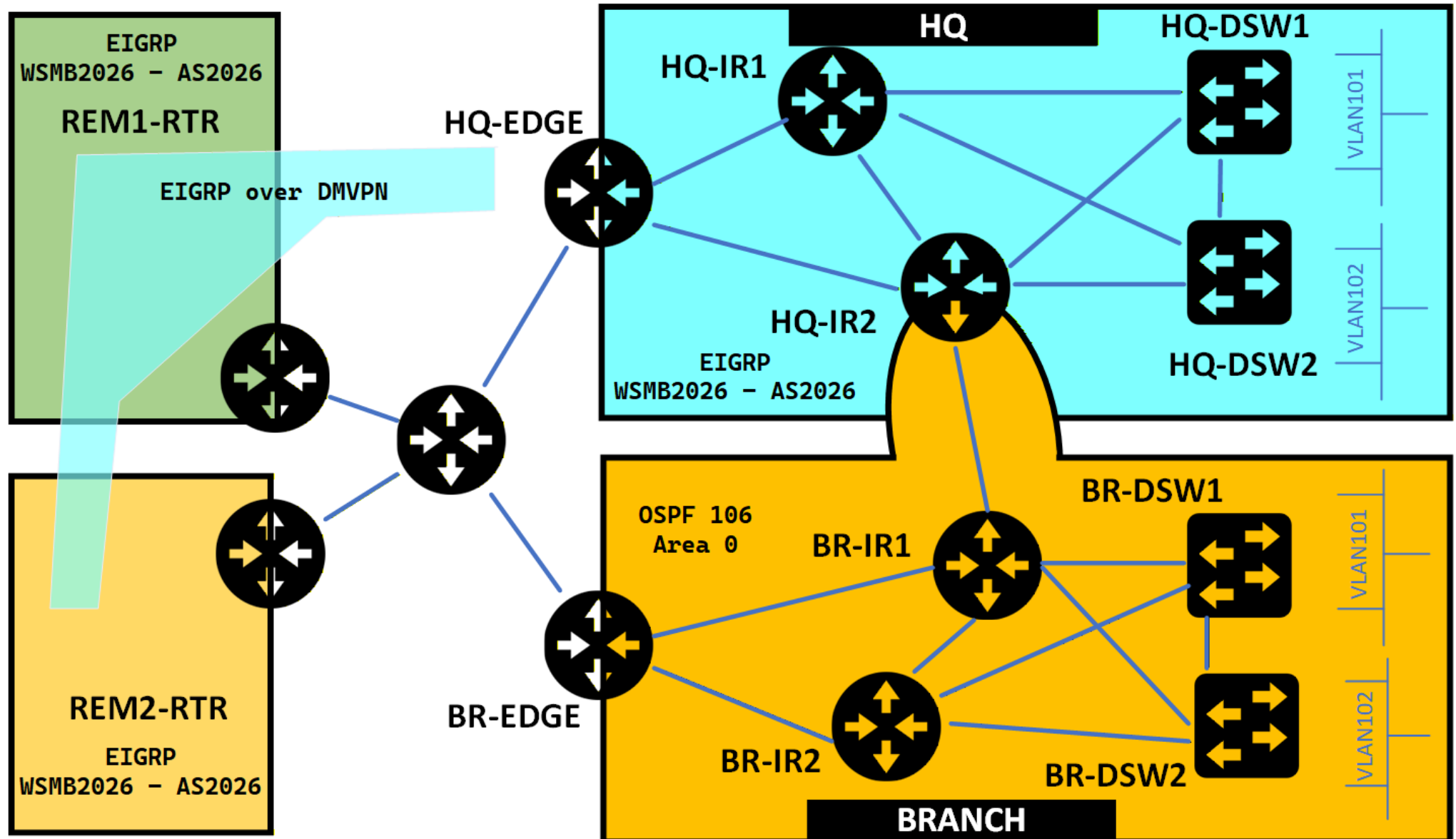
EIGRP OSPF IPv4 Route Redistribution



WSMB2026 Module C - IPv4 Routing Topology



WSMB2026 Module C – IPv6 Routing Topology



Network Topology

